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Pharmacode 6561

# FONKO-CALCIUM FOLINATE
## Solution for injection 50 mg/5 ml

**Composition:**
Each ml contains
Calcium folinate hydrate equivalent to Folinic acid 10 mg

Excipient with known effect:
Calcium folinate 50 mg/5 ml solution for injection contains 16.77 mg of sodium (main component of cooking/table salt) in each 5 ml vial. This is equivalent to 0.8% of the recommended maximum daily dietary intake of sodium for an adult.

**Product Description:**
Clear and yellowish solution. Practically free of foreign matter.
After dilution: clear, colorless solution, practically free of foreign matter.

**Pharmacodynamics:**
Calcium folinate hydrate is a mixture of the diastereoisomers of the 5-formyl derivative of tetrahydrofolic acid (THF). The biologically active compound of the mixture is the (-)-l-isomer, known as citrovorum factor or (-)-folinic acid. Calcium folinate hydrate does not require reduction by the enzyme dihydrofolate reductase in order to participate in reactions utilizing folates as a source of ''one-carbon" moieties.
Administration of calcium folinate hydrate can counteract the therapeutic and toxic effects of folic acid antagonists such as methotrexate, which act by inhibiting dihydrofolate reductase.
In contrast, calcium folinate hydrate can enhance the therapeutic and toxic effects of fluoropyrimidines used in cancer therapy, such as 5-fluorouracil.
Concurrent administration of calcium folinate hydrate does not appear to alter the plasma pharmacokinetics of 5-fluorouracil. 5-fluorouracil is metabolized to fluorodeoxyuridylic acid, which binds to and inhibits the enzyme thymidylate synthase (an enzyme important in DNA repair and replication).
Calcium folinate hydrate is readily converted to another reduced folate, 5, 10-methylenetetrahydrofolate, which acts to stabilize the binding of fluorodeoxyridylic acid to thymidylate synthase and thereby enhances the inhibition of this enzyme.

**Pharmacokinetics:**
***Absorption***
Peak plasma levels of folinic acid are reached, on average 40 minutes and 1.7 hours after intramuscular and oral administration, respectively. The bioavailability of an oral dose is almost the same as an equivalent intramuscular dose.

***Distribution***
Tetrahydrofolic acid and its derivatives are distributed to all body tissues, being concentrated in the liver and found in moderate amounts in the CSF following a 15 mg dose given either orally or intramuscularly, peak serum folate concentrations of 0.268 micrograms/ml and 0.241 micrograms/ml were detected.

***Metabolism***
Calcium folinate hydrate is rapidly and extensively converted to 5-methyl tetrahydrofolate (an active metabolite) *in vivo*, with less extensive conversion resulting from parenteral, as opposed to oral administration.

***Elimination***
Folinic acid is eliminated mainly as 10-formyl tetrahydrofolate and 5, 10-methyl tetrahydrofolate. The metabolites are mainly excreted via the urine (80–90%), with elimination being logarithmic in doses exceeding 1 mg.

**Indications:**
- Calcium folinate hydrate has shown good results in reducing the toxicity and circumventing the effect of folic acid antagonists, if therapeutically desired.
- Use in combination with 5-fluorouracil in the treatment of advanced colorectal carcinoma.
- Calcium folinate hydrate has shown good results in the treatment of certain megaloblastic anemia resulting from folic acid deficiency. This mainly occurs in infants, during pregnancy, in malabsorption syndromes, liver diseases, sprue, and malnutrition. It is not more effective than folic acid for these conditions.

**Recommended Dose:**
Calcium folinate hydrate may be given parenterally by intramuscular injection, intravenous injection, or intravenous infusion. Calcium folinate hydrate should not be administered intrathecally.

***Calcium folinate hydrate rescue after methotrexate therapy***
The recommendations for calcium folinate hydrate rescue are based on a methotrexate dose of 12 to 15 gram/$m^2$ administered by intravenous infusion over 4 hours. Calcium folinate hydrate rescue at a dose of 15 mg (approximately 10 mg/$m^2$) every 6 hours for 10 doses starts 24 hours after the beginning of the methotrexate infusion. In the presence of gastrointestinal toxicity, nausea, or vomiting, calcium folinate hydrate should be administered parenterally.

Serum creatinine and methotrexate levels should be determined at least once daily. Calcium folinate hydrate administration, hydration, and urinary alkalinization (pH of 7.0 or greater) should be continued until the methotrexate level is below $5 \times 10^{-8}$ M (0.05 micromolar). The calcium folinate hydrate dose should be adjusted or calcium folinate hydrate rescue extended based on the following guidelines:

| Guidelines for calcium folinate hydrate dosage and administration | | |
|---|---|---|
| **Clinical situation** | **Laboratory findings** | **Calcium folinate hydrate dosage and duration** |
| Normal methotrexate elimination | Serum methotrexate level approximately 10 micromolar at 24 hours after administration, 1 micromolar at 48 hours, and less than 0.2 micromolar at 72 hours | 15 mg IM or IV q 6 hours for 60 hours (10 doses starting at 24 hours after start of methotrexate infusion) |
| Delayed late methotrexate elimination | Serum methotrexate level remaining above 0.2 micromolar at 72 hours, and more than 0.05 micromolar at 96 hours after administration | Continue 15 mg IM or IV q 6 hours, until methotrexate level is less than 0.05 micromolar |

Patients who experience delayed methotrexate elimination are likely to develop reversible renal failure. These patients require continuing hydration, urinary alkalinization, and close monitoring of fluid and electrolyte status, until the serum methotrexate level has fallen to below 0.05 micromolar and the renal failure has resolved.

Some patients will have abnormalities in methotrexate elimination or renal function following methotrexate administration, which are significant but less severe than the abnormalities described above. If these abnormalities are associated with significant clinical toxicity, calcium folinate hydrate rescue should be extended for an additional 24 hours (total of 14 doses over 84 hours) in subsequent courses of therapy. The possibility that the patient is taking other medications which interact with methotrexate (e.g., medications which may interfere with methotrexate elimination or binding to serum albumin) should always be reconsidered when laboratory abnormalities or clinical toxicities are observed.

***Infusion direction***
Calcium folinate hydrate may be diluted for infusion with 0.9% sodium chloride solution or 5% dextrose solution. Caution: contains no preservative. If intended for multiple dose use, reconstitute with bacteriostatic water for injection containing benzyl alcohol. The pH of the solution is approximately 7.5. Reconstituted solutions should not be kept for more than 24 hours and should be stored between 2–8°C.

Calcium folinate hydrate should not be mixed in the same infusion as 5-fluorouracil as a precipitate may form.

***Impaired methotrexate elimination or inadvertent overdosage***
Calcium folinate hydrate rescue should begin as soon as possible after an inadvertent overdosage and within 24 hours of methotrexate administration when there is delayed excretion. Calcium folinate hydrate 10 mg/$m^2$ should be administered IV or IM every 6 hours until the serum methotrexate level is less than $10^{-8}$ M. Serum creatinine and methotrexate levels should be determined at 24-hour intervals. If the 24 hours serum creatinine has increased 50% over baseline or if the 24-hour methotrexate level is greater than $5 \times 10^{-6}$ M or the 48-hour level is greater than $9 \times 10^{-7}$ M, the dose of calcium folinate hydrate should be increased to 100 mg/$m^2$ IV every 3 hours until the methotrexate level is less than $10^{-8}$ M.

Hydration (3 l/day) and urinary alkalinization with sodium bicarbonate solution should be employed concomitantly. The bicarbonate dose should be adjusted to maintain the urine pH at 7.0 or greater.
Note: high dose methotrexate therapy should only be administered by qualified specialists and in hospitals where the necessary facilities are available. Recent published literature should be consulted for details at all times.

***Treatment of pyrimethamine overdosage***
The dosage of pyrimethamine in treating toxoplasmosis is 10 to 20 times its dosage for malaria and approaches the toxic level. Since calcium folinate hydrate is not utilized by protozoa, it can be given simultaneously without impairing the effectiveness of therapy. The usual dosage is 3 to 9 mg per day intramuscularly for three days or until the platelet and leukocyte counts have reached safe levels.

***Advanced colorectal cancer***
Either of the following two regimens is recommended:
1. Calcium folinate hydrate is administered at 200 mg/$m^2$ by slow intravenous injection over a minimum of 3 minutes, followed by 5-fluorouracil at 370 mg/$m^2$ by intravenous injection.
2. Calcium folinate hydrate is administered at 20 mg/$m^2$ by intravenous injection followed by 5-fluorouracil at 425 mg/$m^2$ by intravenous injection.

5-Fluorouracil and calcium folinate hydrate should be administered separately to avoid the formation of a precipitate.

Treatment is repeated daily for five days. This five-day treatment course may be repeated at 4-week (28-day) intervals, for 2 courses and then repeated at 4 to 5 week (28 to 35 day) intervals provided that the patient has completely recovered from the toxic effects of the prior treatment course.

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In subsequent treatment course, the dosage of 5-fluorouracil should be adjusted based on patient tolerance of the prior treatment course.

The daily dosage of 5-fluorouracil should be reduced by 20% for patients who experienced moderate hematologic or gastrointestinal toxicity in the prior treatment course, and by 30% for patients who experienced severe toxicity. For patients who experienced no toxicity in the prior treatment course, 5-fluorouracil dosage may be increased by 10%. Calcium folinate hydrate dosages are not adjusted for toxicity.

Several other doses and schedules of calcium folinate hydrate/5-fluorouracil therapy have also been evaluated in patients with advanced colorectal cancer; some of these alternative regimens may also have efficacy in the treatment of this disease. However, further clinical research will be required to confirm the safety and effectiveness of these alternative calcium folinate hydrate/5- fluorouracil treatment regimens.

***Megaloblastic anemia due to folic acid deficiency***
Up to 1 mg daily. There is no evidence that doses greater than 1 mg/day have greater efficacy than those of 1 mg; additionally, loss of folate in urine becomes roughly logarithmic as the amount administered exceeds 1 mg.

Parenteral products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

**Route of Administration:**
Intravenous and intramuscular.

**Contraindications:**
Calcium folinate hydrate is improper therapy for pernicious anemia and other megaloblastic anemia secondary to the lack of vitamin B12. When treating these conditions with calcium folinate hydrate, hematological remission may occur, but neurological manifestations are likely to progress.

**Warnings and Precautions:**
- Calcium folinate hydrate is not suitable for the treatment of pernicious anemia and other anemia resulting from lack of vitamin B12. Hematological remissions may occur, while the neurological manifestations remain progressive. Simultaneous therapy with a folic acid antagonist and calcium folinate hydrate is not recommended because the effect of the folic acid antagonist is either reduced or completely inhibited.
- Because of the $Ca^{++}$ content of the calcium folinate hydrate injections, no more than 16 ml of the 10 mg/ml formulations (160 mg of calcium folinate hydrate) should be injected intravenously per minute.
- Calcium folinate hydrate may enhance the toxicity of fluorouracil. Deaths from severe enterocolitis, diarrhea, and dehydration have been reported in elderly patients receiving calcium folinate hydrate and fluorouracil. Concomitant granulocytopenia and fever were present in some, but not all, of the patients.
- Seizures and/or syncope have been reported rarely in cancer patients receiving calcium folinate hydrate, usually in association with fluoropyrimidine administration, and most commonly in those with CNS metastases. Since three patients had recurrent neurological symptoms on rechallenge with calcium folinate hydrate further treatment with calcium folinate hydrate is not recommended in these circumstances.
- Calcium folinate hydrate has no effect on nonhematological toxicities of methotrexate, such as the nephrotoxicity resulting from drug and/or metabolite precipitation in the kidney.
- Calcium folinate hydrate should only be used with folic acid antagonists, e.g., methotrexate or fluoropyrimidines, e.g., 5-fluorouracil, under the direct supervision of a clinician experienced in the use of cancer chemotherapeutic agents.
- Combination therapy with 5-fluorouracil:
  Calcium folinate hydrate enhances the toxicity of 5-fluorouracil. When these drugs are administered concurrently in the palliative therapy of advanced colorectal cancer, the dosage of 5-fluorouracil should be reduced accordingly.
  Although the toxicities observed in patients treated with the combination of calcium folinate hydrate followed by 5-fluorouracil are qualitatively similar to those observed in patients treated with 5-fluorouracil alone, gastrointestinal toxicities (particularly stomatitis and diarrhea) are observed more commonly, may be more severe and of prolonged duration in patients treated with the combination.
  Combination therapy with calcium folinate hydrate/5-fluorouracil must not be initiated or continued in patients who have symptoms of gastrointestinal toxicity of any severity, until those symptoms have completely resolved. Patients with diarrhea must be monitored with particular care until the diarrhea has resolved, as rapid clinical deterioration leading to death can occur.
  Particular care should be taken when treating elderly or debilitated colorectal cancer patients with calcium folinate hydrate/5-fluorouracil, as these patients may be at increased risk of severe toxicity, particularly severe gastrointestinal toxicity.
- Laboratory test - combination therapy with 5-fluorouracil:
  Patients being treated with the calcium folinate hydrate/5-fluorouracil combination should have a CBC with differential and platelets prior to each treatment. During the first two courses a CBC with differential and platelets has to be repeated weekly and thereafter, once each cycle at the time of anticipated WBC nadir. Electrolytes and liver function tests should be performed prior to each treatment for the first three cycles then prior to every other cycle. Dosage modifications of 5-fluorouracil should be instituted as follows, based on the most severe toxicities:

| Diarrhea and/or stomatitis | WBC/mm³ nadir | Platelets/mm³ nadir | 5-FU dose |
|---|---|---|---|
| Moderate | 1,000–1,900 | 25–75,000 | Decrease 20% |
| Severe | <1,000 | <25,000 | Decrease 30% |

  If no toxicity occurs, the 5-fluorouracil dose may be increased 10%. Treatment should be deferred until WBC's are 4,000/mm³. If blood counts do not reach these levels within two weeks, treatment should be discontinued. Patients should be followed up with physical examination prior to each treatment course and appropriate radiological examination as needed. Treatment should be discontinued when there is clear evidence of tumor progression.

***Pediatric use***
Folic acid in large amounts may counteract the antiepileptic effect of phenobarbital, phenytoin, and primidone and increase the frequency of seizures in susceptible pediatric patients.

**Interactions with Other Medicaments:**
- Folic acid in large amounts may counteract the antiepileptic effect of phenobarbital, phenytoin, and primidone and increase the frequency of seizures in susceptible children.
- High intravenous or intramuscular doses of calcium folinate hydrate may reduce efficacy of intrathecally administered methotrexate.
- Calcium folinate hydrate may enhance the toxicity of fluorouracil.

**Pregnancy and Lactation:**
***Pregnancy***
Teratogenic effects: pregnancy category C.
Adequate animal reproduction studies have not been conducted with calcium folinate hydrate. It is also not known whether calcium folinate hydrate can cause fetal harm when administered to a pregnant woman or can affect reproduction capacity. Calcium folinate hydrate should be given to a pregnant woman only if clearly needed.

***Lactation***
It is not known whether calcium folinate hydrate is excreted into human milk. Because many drugs are excreted in human milk, caution should be exercised when calcium folinate hydrate is administered to a nursing mother.

**Adverse Effects:**
Allergic sensitization, including anaphylactoid reactions and urticaria, has been reported following both oral and parenteral administration of folic acid. Nausea and vomiting with very high doses of calcium folinate hydrate have been reported. Seizures and/or syncope have been reported rarely in cancer patients receiving calcium folinate hydrate, usually in association with fluoropyrimidine administration.

**Overdose and Treatment:**
Excessive amounts of calcium folinate hydrate may nullify the chemotherapeutic effect of folic acid antagonists.

**Incompatibilities:**
Folic acid has been reported to be incompatible with droperidol and phosphonosulphate.

**Presentation and Registration Number:**
Box, 1 vial x 5 ml; MAL24036031AZ

**ON MEDICAL PRESCRIPTION ONLY.**

**STORE AT TEMPERATURES BETWEEN 2–8ºC. PROTECT FROM HEAT AND LIGHT.**

**DO NOT FREEZE.**

**KEEP MEDICINES OUT OF REACH OF CHILDREN.**

**AFTER DILUTION IN 0.9% SODIUM CHLORIDE SOLUTION OR 5% DEXTROSE SOLUTION, THE SOLUTION IS STABLE FOR 24 HOURS WHEN STORED BETWEEN 2–8ºC.**

**FROM MICROBIOLOGICAL POINT OF VIEW, THE PRODUCT SHOULD BE USED IMMEDIATELY. IF NOT USED IMMEDIATELY, IN-USE STORAGE TIMES AND CONDITIONS PRIOR TO USE ARE THE RESPONSIBILITY OF THE USER AND WOULD NOT BE LONGER THAN 24 HOURS AT 2–8ºC.**

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Date of revision: 10 January 2024

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